

Advanced Audio Processing for VoNR

A Software Engineer's Blueprint for AEC and NR Implementations under 5G Constraints.



The Engineering Challenge: VoLTE vs. VoNR Constraints

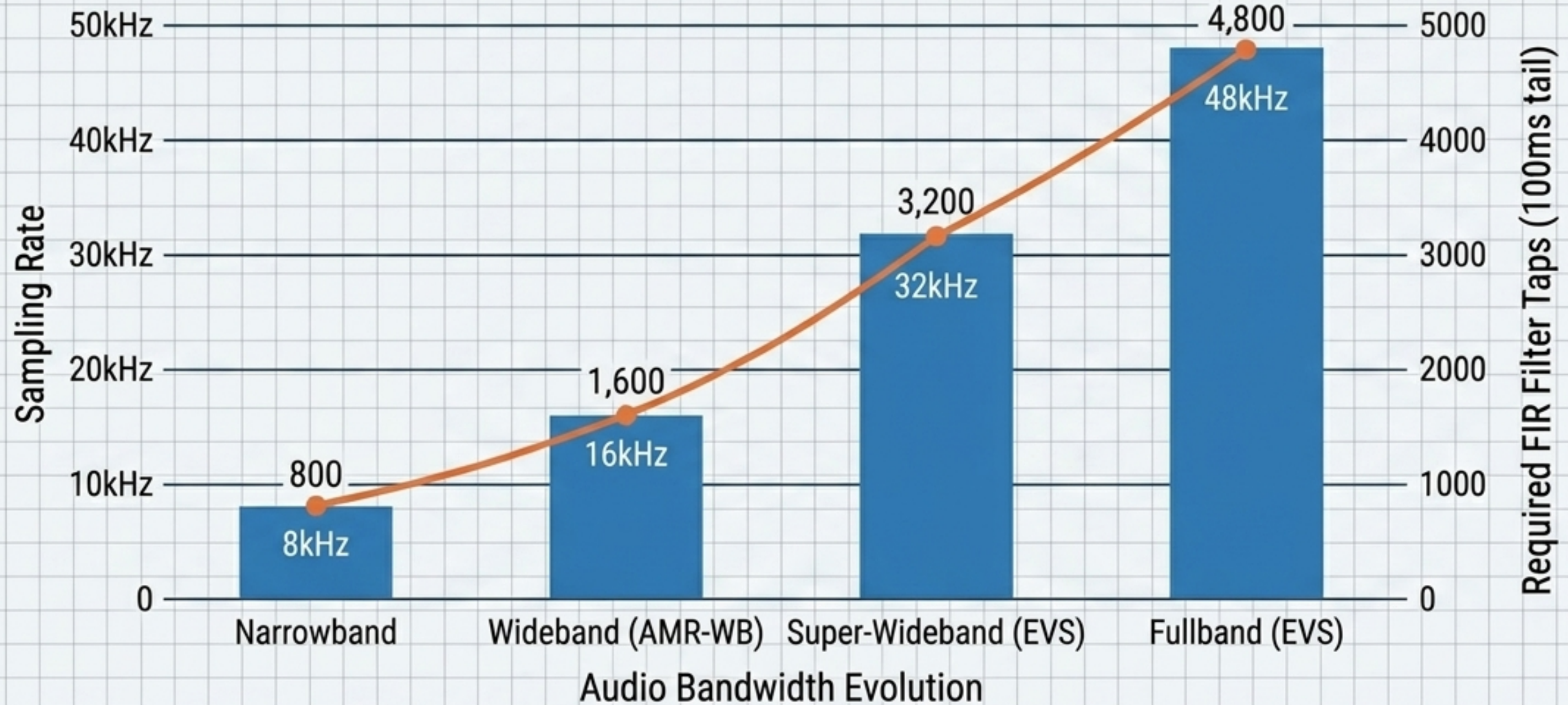
	VoLTE (4G)	VoNR (5G)
Core Network:	EPC	5GC
Bearer / QoS:	QCI 1	5QI 1 (+ Interactive DC)
Call Setup:	2-3 Seconds	< 1 Second
Target Codecs:	AMR-WB / EVS	EVS / IVAS
Mouth-to-Ear Latency:	100-150ms	< 50ms (Target)



Engineer's Takeaway:

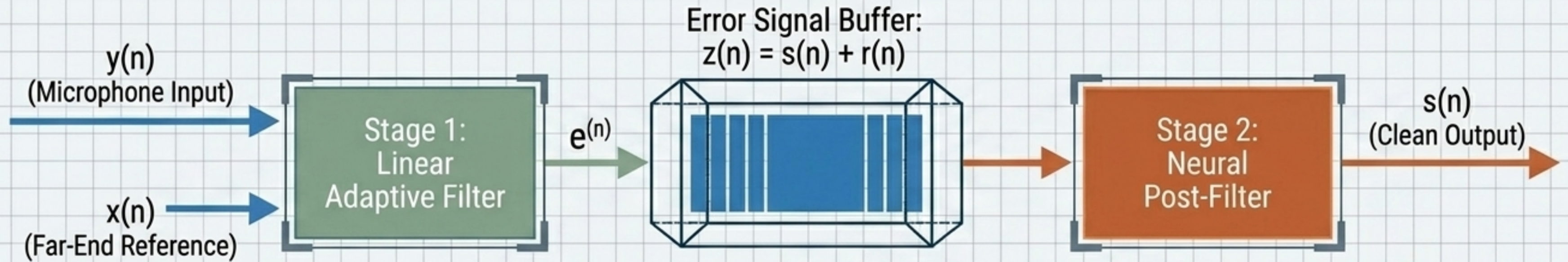
The drop from 150ms to **<50ms** total latency dictates that DSP algorithms must execute in **sub-millisecond windows**. Legacy **STFT-based buffering (16-32ms)** severely compromises this budget.

The EVS Codec Paradigm Shift & Compute Explosion



Processing a 48kHz Fullband EVS signal requires **4,800 discrete tap calculations** per audio channel, per 100ms tail. This exponentially scales the required **Millions of Instructions Per Second (MIPS)**, mandating wide-SIMD processing architectures.

Architecture Overview: The Hybrid DSP-DNN Pipeline



Stage 1: Linear AEC (DSP)

- Mathematically efficient at tracking dynamic room impulse responses.
- Suppresses the bulk of the linear acoustic echo.

Stage 2: Neural Post-Filter (DNN)

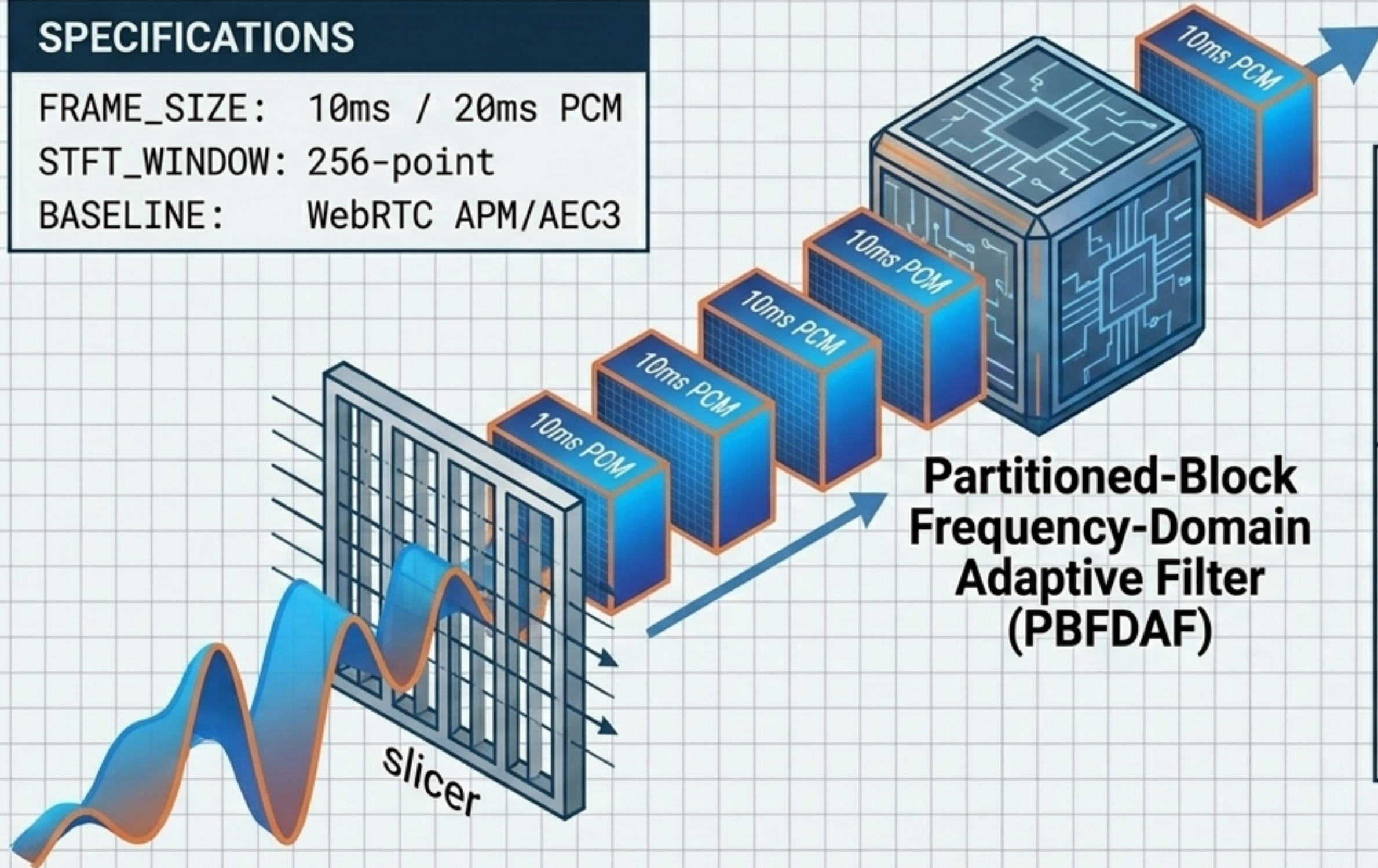
- Operates ONLY on the residual error signal.
- Trained exclusively to suppress non-linear residual echo and ambient noise.

Pure End-to-End Deep Learning models require massive parameter counts. Partitioning the linear workload saves critical **MIPS**.

Module 1: Linear AEC (PBFDAF Implementation)

SPECIFICATIONS

FRAME_SIZE: 10ms / 20ms PCM
STFT_WINDOW: 256-point
BASELINE: WebRTC APM/AEC3



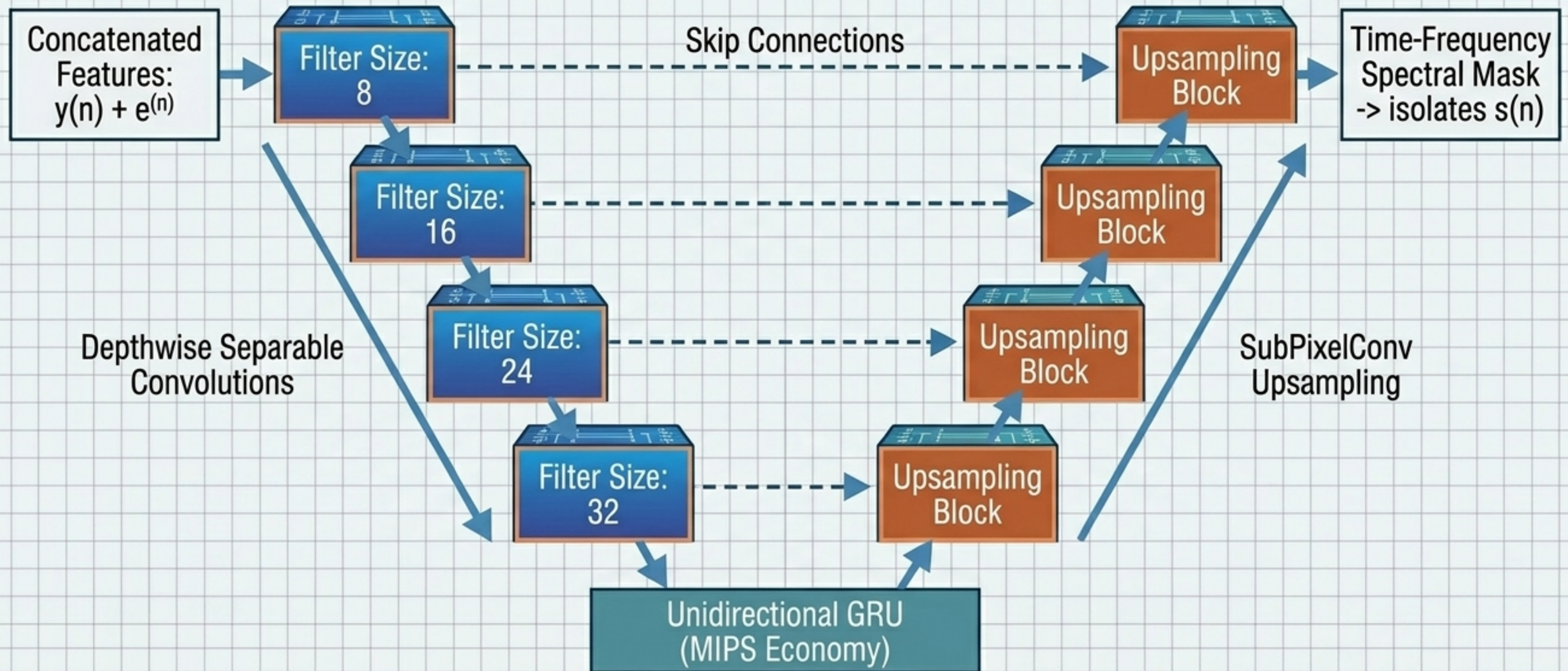
Implementation Criticals

Jitter Buffer Management (JBM): Must tightly integrate with the 5G network stack.

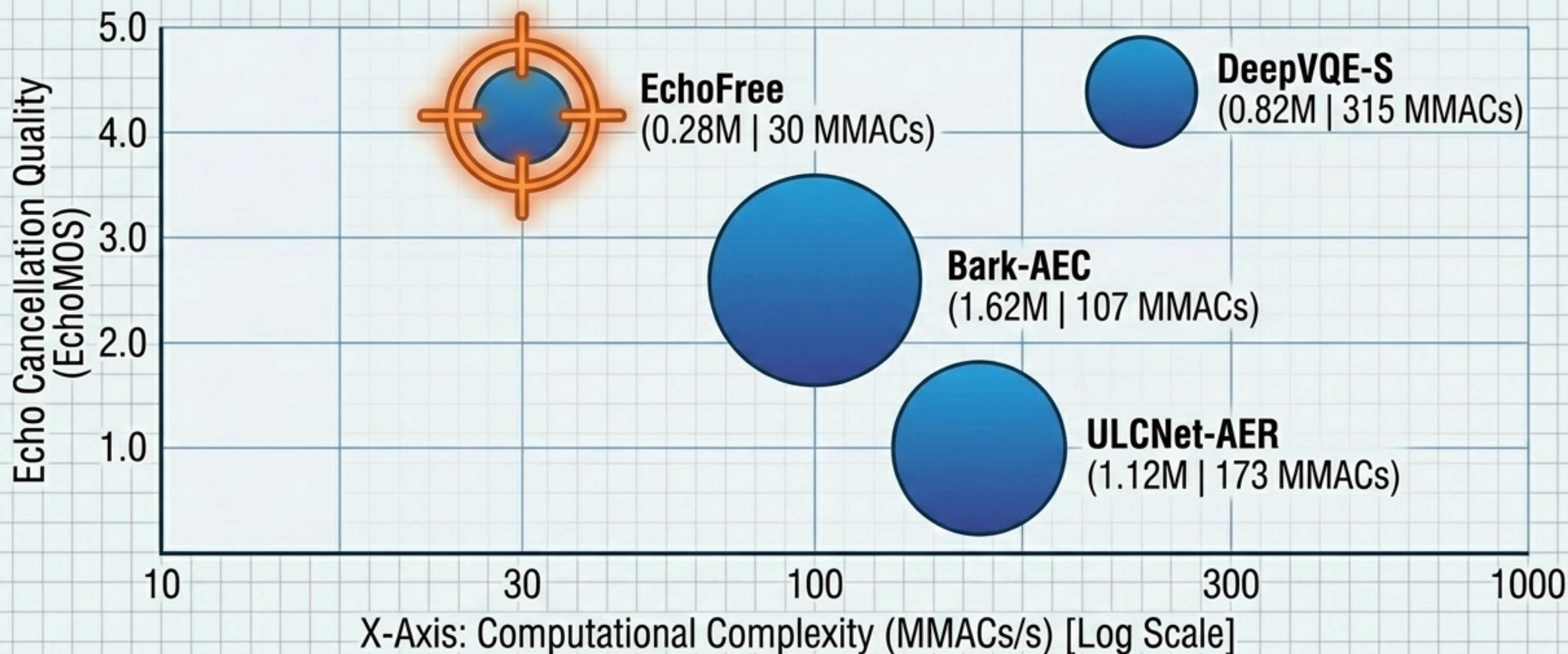
Clock Drift: APM requires constant estimation of sound-card delay and skew to prevent echo path model divergence.

EPS Fallback: Algorithm must survive 20-30ms network-induced audio clock glitches during 5G-to-4G handovers without blasting residual echo.

Module 2: Neural Post-Filtering (U-Net Architecture)

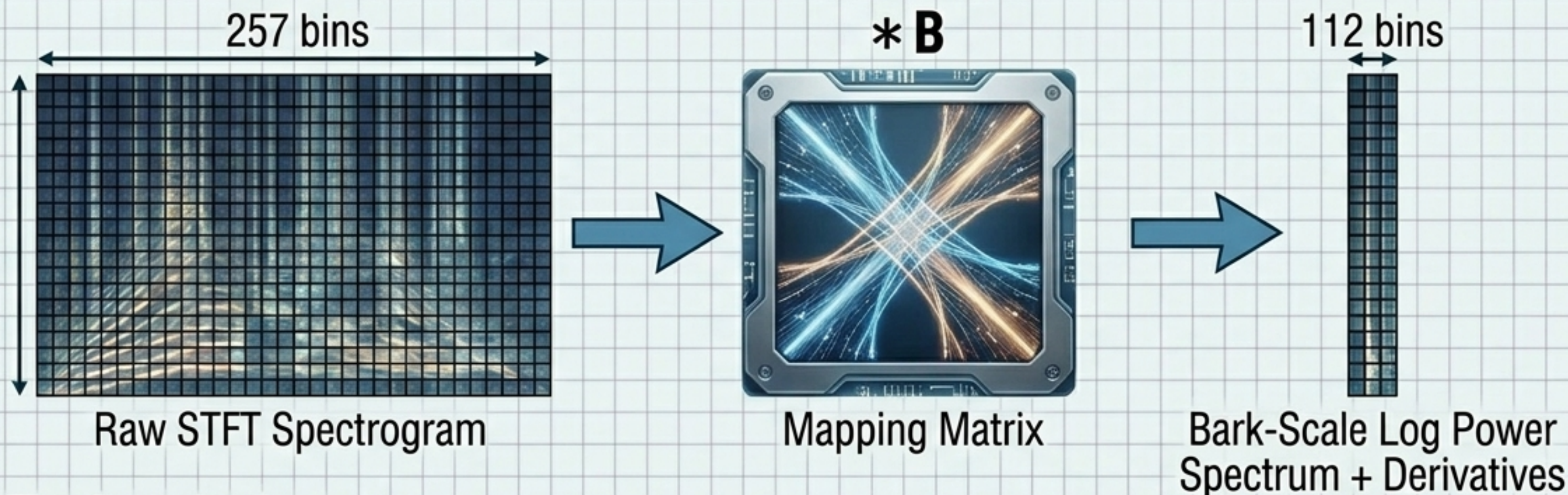


SOTA Model Profiling: The Efficiency Trade-off



Conclusion: EchoFree establishes a superior trade-off, achieving SOTA performance within a 30 MMACs budget—ideal for Cortex-M and battery-constrained IoT endpoints.

Feature Engineering: Bark-Scale Spectral Compression

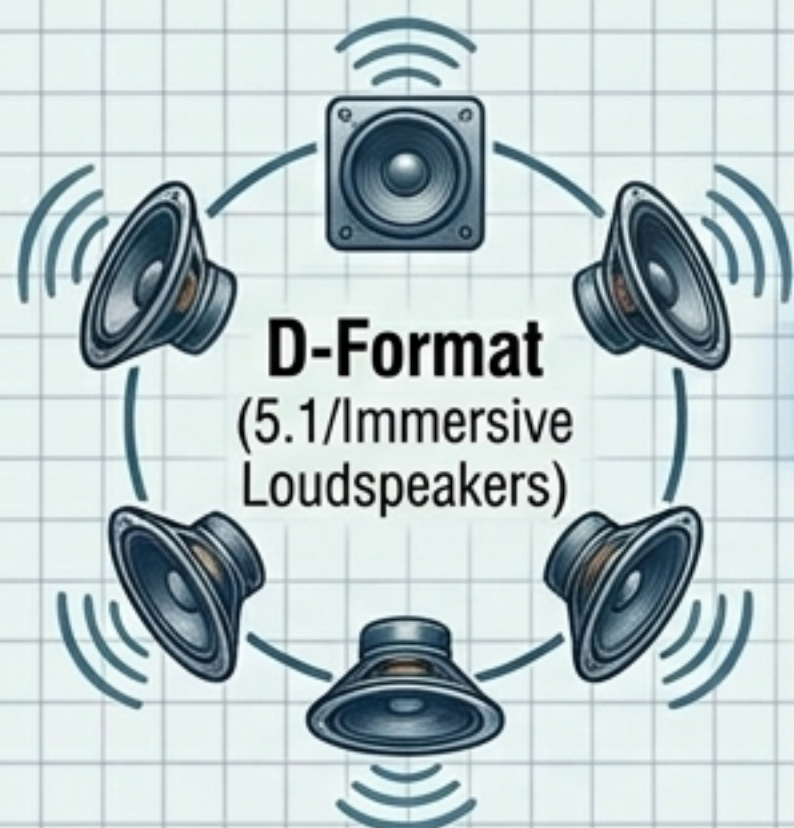


- **The Optimization:**

- Raw STFT yields 257 frequency bins at 16kHz—too heavy for first-layer neural processing.
- Multiplied by mapping matrix to extract 100 Bark-scale filters.
- First- and second-order derivatives added for prior knowledge.

Perceptually-motivated compression drastically reduces VRAM and MAC operations in the neural encoder layer while preserving human-speech fidelity.

Advanced VoNR+ Use Case: Surround Sound AEC



Fails: Highly correlated signals cause non-unique solutions.



B-Format
(Ambisonics Sphere)

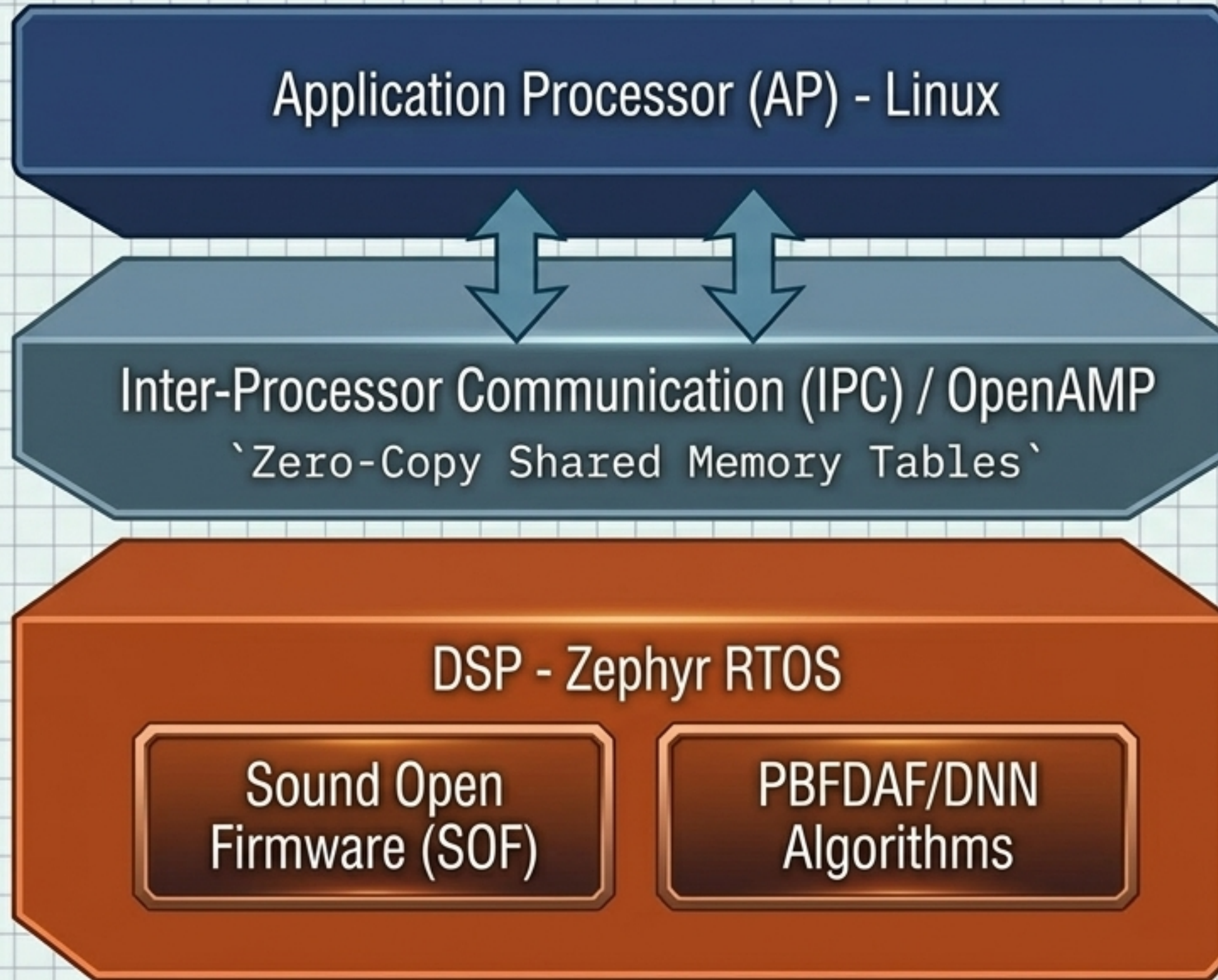


Feed compressed complex spectra directly to network.



Outcome:
B-Format routing achieves superior generalization against mismatched physical loudspeaker layouts and drastically improves Echo Return Loss Enhancement (ERLE).

Firmware Abstraction: SOF, RTOS, and Zero-Copy IPC



Inter-Processor Communication

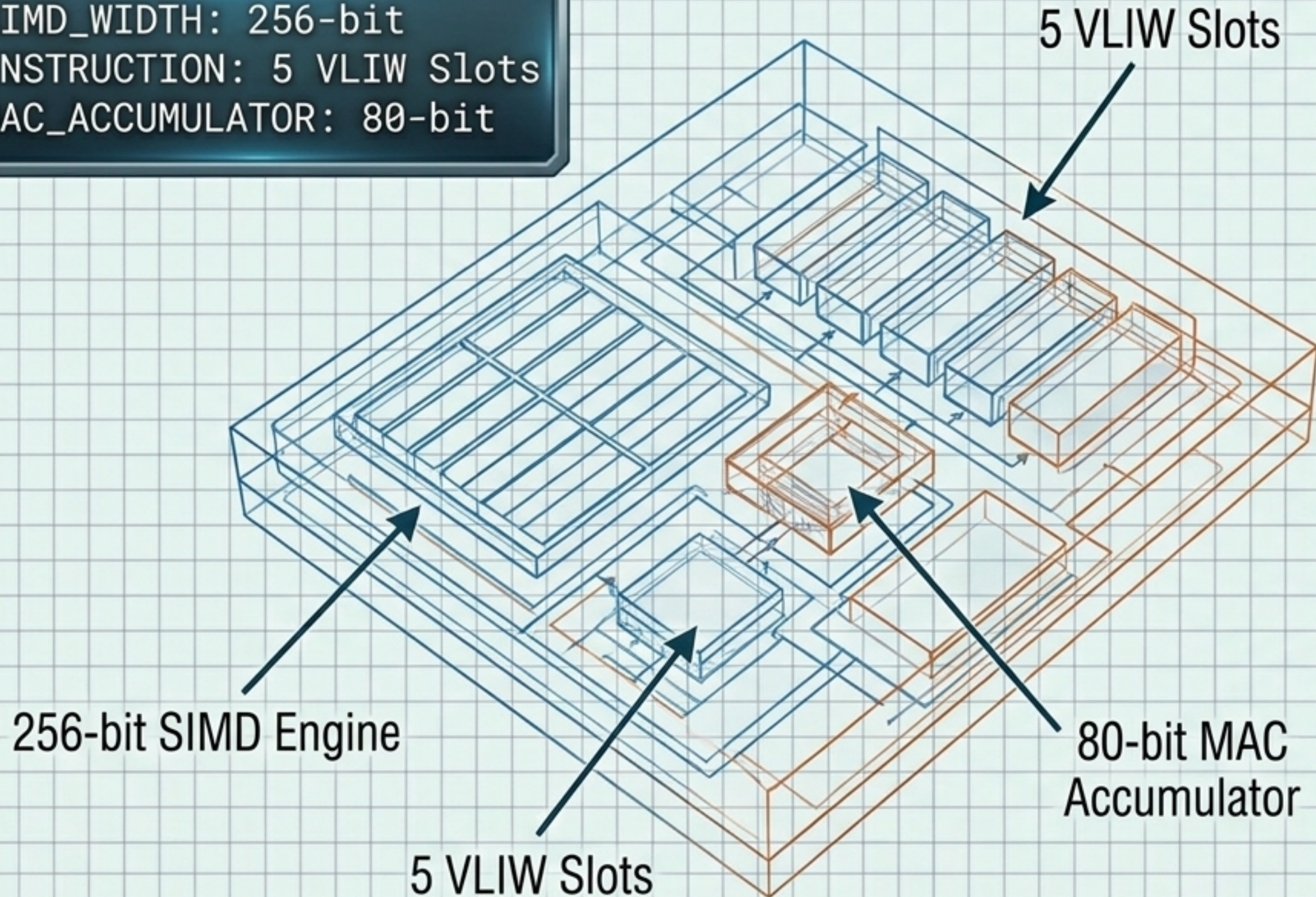
Heterogeneous cores maintain different memory maps. Generic address translation prevents massive latency penalties from copying audio buffers between AP and DSP.

Sound Open Firmware (SOF)

Abstracts hardware complexities across silicon generations. Ensures complex neural AEC pipelines can be ported without total firmware rewrites.

Hardware Target 1: Cadence Tensilica HiFi iQ

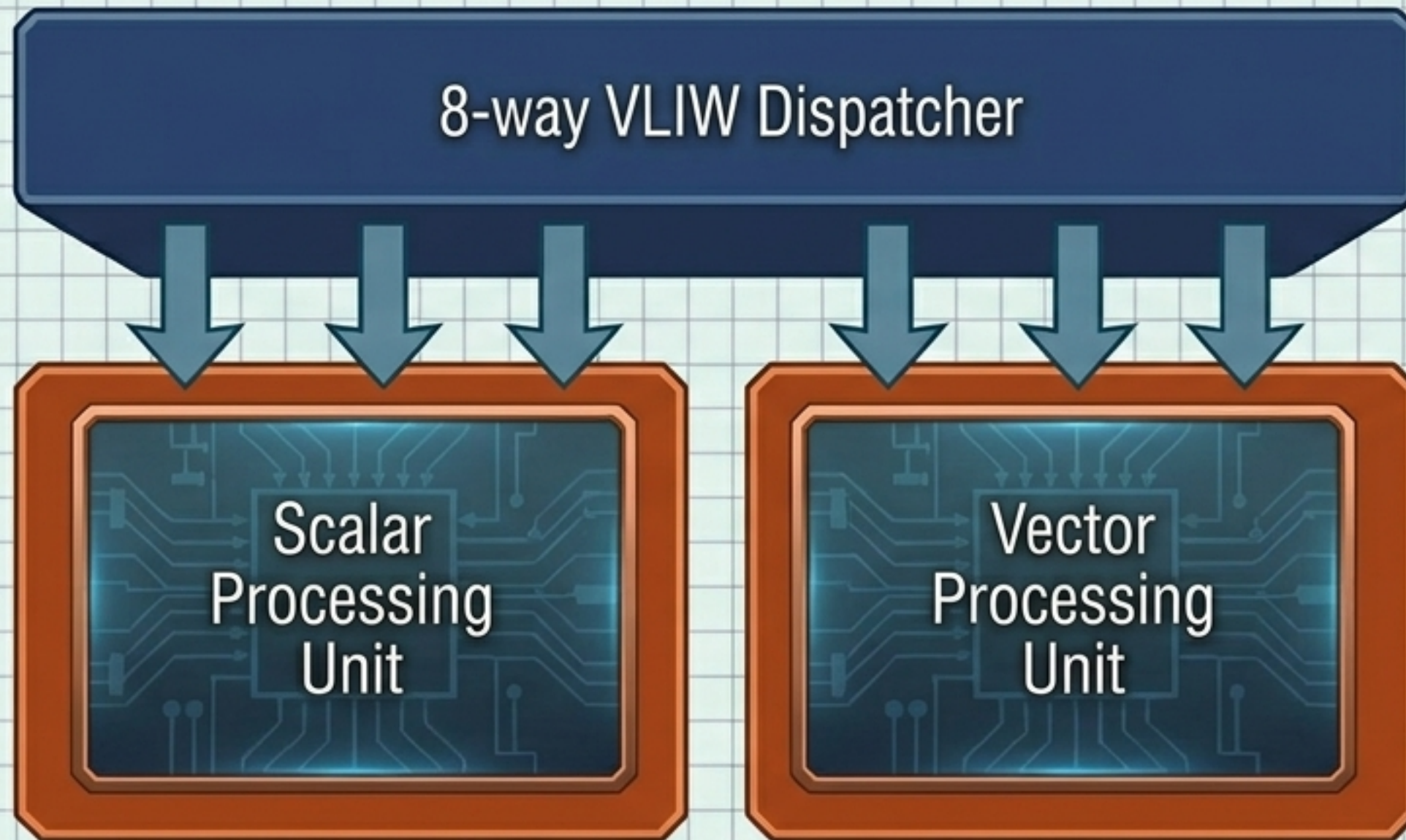
SIMD_WIDTH: 256-bit
INSTRUCTION: 5 VLIW Slots
MAC_ACCUMULATOR: 80-bit



Software Optimization:

- Native hardware execution of FP8 and BF16 for heavily quantized NNs.
- Utilize Cadence NeuralNet Library (NNLIB) paired with TensorFlow Lite for Microcontrollers (TFLM).
- Cycle count improvements up to 458x for Conv2D operators vs reference software.

Hardware Target 2: CEVA SensPro2 & ClearVox



PERFORMANCE: 0.2 to 2 TOPS
DATAPATH: 8-way VLIW
AUDIO_RES: 16kHz / 16-bit



Ultra-Low Power:
Days/Weeks Uptime

Software Optimization

- Implement Hardware Loop Buffers and Branch Target Buffers to drastically reduce instruction fetch power.
- ClearVox ENC executes pure edge AI noise reduction using a single mic input, eliminating cloud latency for "always-on" IoT applications.

Hardware Target 3: NXP i.MX & VoiceSeeker

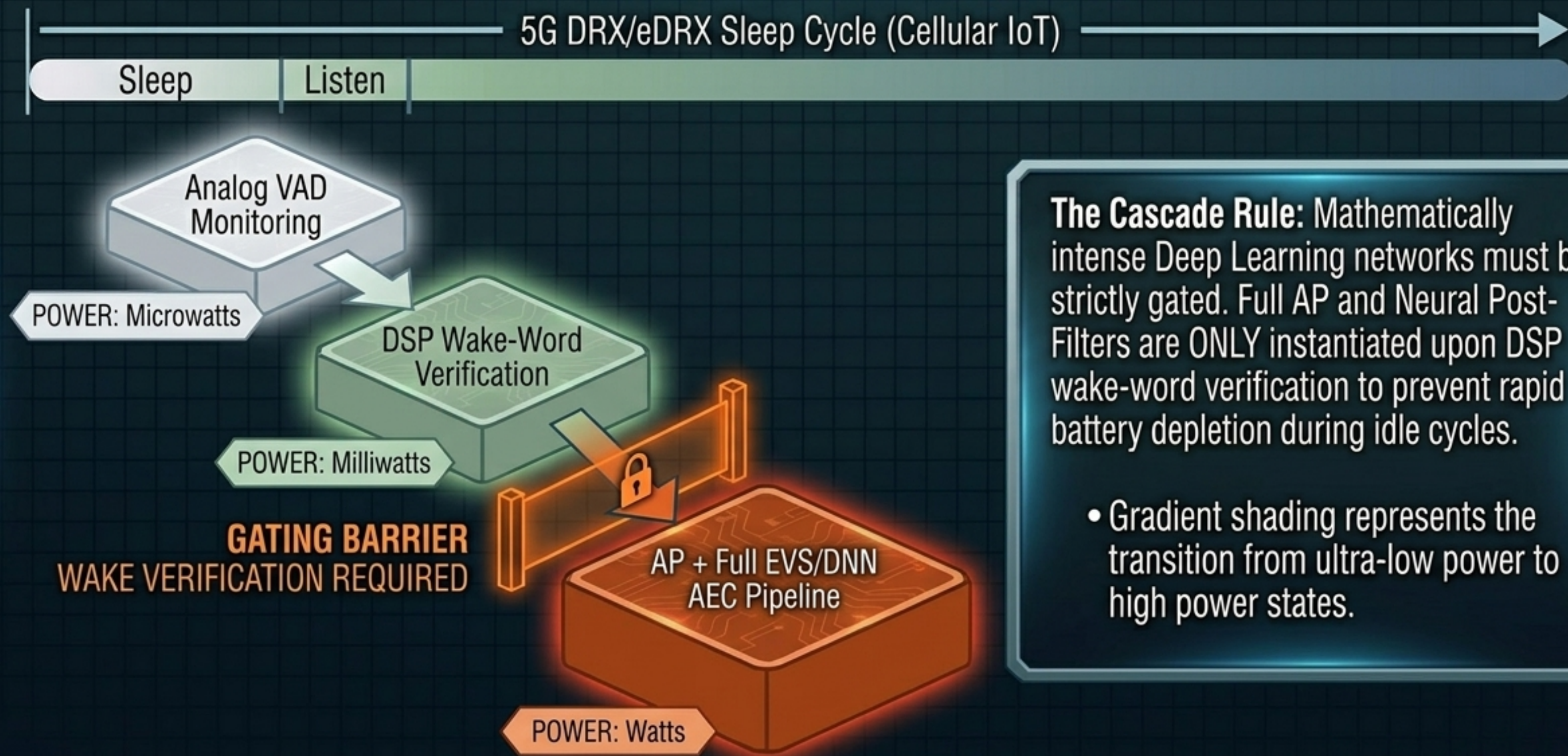


The Software Advantage: Algorithm dynamically adapts at runtime to mic spacing (2cm to 8cm) without strict hardware tolerances.

Performance Profile:

- Latency: ~14ms total with AEC active.
- Resource Scale: 14MHz/60kB (2-mic) -> 320MHz/290kB (3-mic + speakers + AEC).
- Leverages elQ Neutron NPU for transient background noise removal.

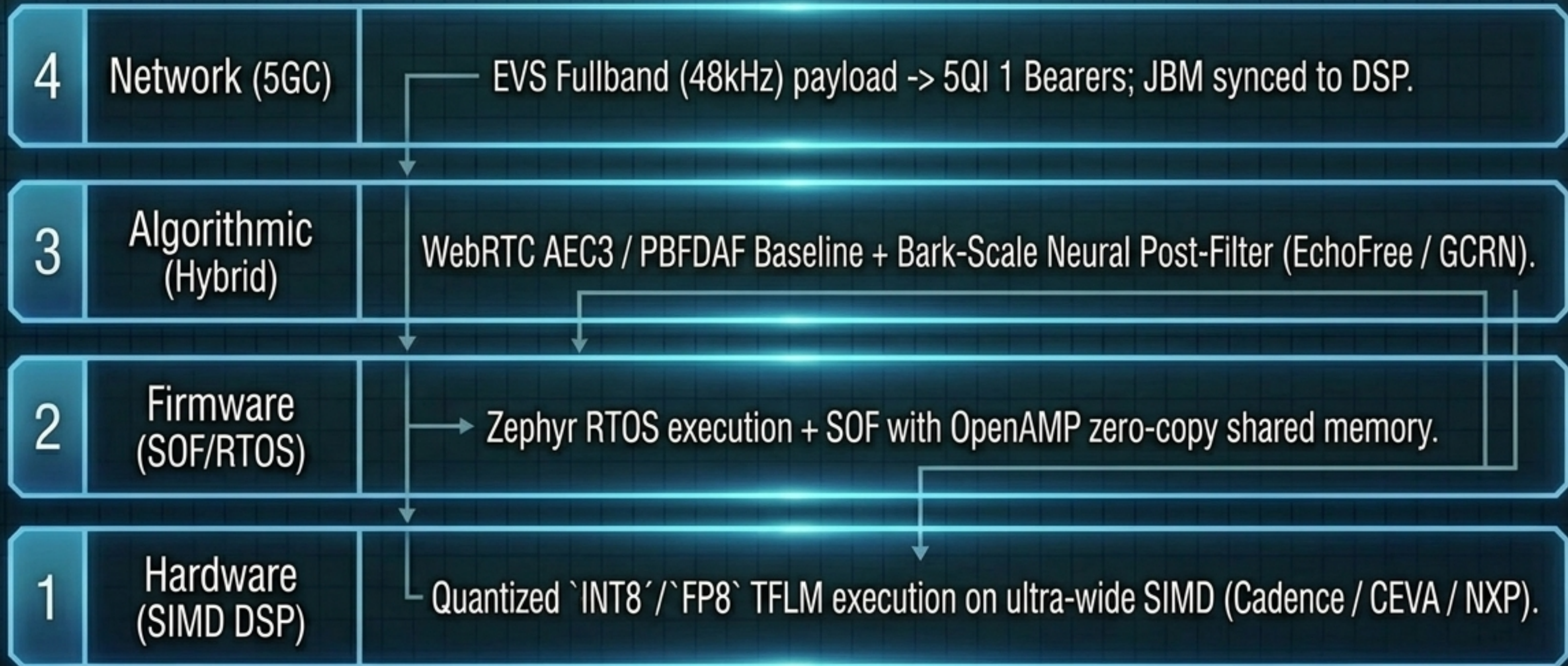
System-Level Power Management & Network States



The Cascade Rule: Mathematically intense Deep Learning networks must be strictly gated. Full AP and Neural Post-Filters are ONLY instantiated upon DSP wake-word verification to prevent rapid battery depletion during idle cycles.

- Gradient shading represents the transition from ultra-low power to high power states.

Synthesis: The VoNR Audio Software Master Plan



The optimal 5G voice stack does not replace DSP with AI; it uses hardware-accelerated, quantized AI to clean what linear DSP leaves behind, operating strictly within a 50ms total latency budget.